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Scaling of earthquake waiting time distributions in northern Chile

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SUMMARY

In this study, we examine the scaling of earthquake waiting time distributions in Northern Chile. For this purpose, we analysed 7-yr high spatial resolution and low completeness magnitude IPOC seismic catalogue and 43-yr USGS catalogue. A unified earthquake moment–space–time scaling relation is empirically evaluated by calculating waiting times for different ranges of magnitude and epicentral area linear dimension and estimating the associated scaling coefficients, β analogue to b-value and γ, the correlation fractal dimension. We find a scaling function that can be characterized with three distinct regions, regions whose behaviour depends on whether seismicity is in the coastal area or at intermediate depth. Moreover, high resolution localizations from the IPOC catalogue allows us to further observe differences: the lower plane seismicity of the double seismic zone behaves just like that at intermediate depth, while the crustal upper plate and the upper plane from the double seismic zone seismicity behaves more like interface seismicity. Thus, the earthquake waiting time distribution primarily depends on whether seismicity is located in a crust-on-crust or in a crust–mantle system: having, respectively, high/low correlated behaviour in the short scale region, non-exponential/exponential decay in the transition middle region and, in all cases, long-term clustering with a slower than exponential decay in the long scale.

Key words: Persistence, memory, correlations, clustering; Earthquake hazards; Seismicity and tectonics; Subduction zone processes.

1 INTRODUCTION

Earthquakes are a recurring phenomenon in the Earth's lithosphere. Regardless of the seismically active zone where an earthquake has been observed, it is to be expected that after some waiting time τ another earthquake will be observed in its vicinity.

Quantifying earthquake waiting time probability distribution D is an important step towards hazard estimation. A way to do this is by exploiting the similarity properties of the D distribution as a function of seismic moment M, or with earthquake size S as initially proposed by Christensen *et al.* (2002) and Corral (2003), and with the epicentral area linear dimension L. Then, we assume that:

$$D \propto M^{-\beta} L^{\gamma} \Psi(c M^{-\beta} L^{\gamma} \tau), \quad (1)$$

where Ψ is a scaling function, c is a constant and β, γ are scaling exponents. β is related to the seismic moment power-law distribution (and hence, to the Gutenberg–Richter b-value) and γ to the epicentre power-law (fractal) spatial distribution. This is a particular case of incomplete similarity in the governing parameters M and L through a power-law scaling (Barenblatt 2003). This relation produces a collapse of different waiting time probability distributions given different seismic moment and epicentral area linear dimension to a common scaling function Ψ, and it was named Unified Scaling Relation for Earthquakes (USLE; Christensen *et al.* 2002). The latter approach succeeded in visualizing the similarity of seismicity temporal distribution, and derived applications followed (e.g. Baiesi & Paczuski 2004; Zaliapin *et al.* 2008), but there is still space to explore consequences for seismic hazard and to acknowledge the depending of the scaling on the tectonic context. Although the latter aspect was not at first considered, Davidsen & Goltz (2004) and Corral (2004) found different scaling functions in seismic catalogues corresponding to different tectonic zones.

With methodologies based on the scaling and the clustering of seismicity, regional and global studies have found several differences between different tectonic contexts. For example, Hainzl *et al.* (2019) noted that aftershock productivity decreases with increasing distance from the trench along the slab and associated it to decreasing interplate coupling. Cabrera *et al.* (2021), on the other hand, noted a decreasing aftershock productivity with distance from slab and associated it with local isotherm, which hints that a depth-dependent influence of heat flow may be in place, such us the one influencing side-to-side clustering properties on global shallow ≤70 km seismicity (Zaliapin & Ben-Zion 2016). On a global analysis, Dascher-Cousineau *et al.* (2020) found that aftershock productivity is inversely correlated with hypocentral depth and concluded that volumetric abundance of nearby faults plays a key role in the clustering properties. Could those observations be related to different forms of the earthquake waiting time scaling function for geometrical cluster groups inside the same subduction zone?

Here we explore the hypothesis that differences in the distributions of earthquake waiting times can be observed within the same forearc region, say between coastal seismicity—situated within 200 km from the trench—and intermediate-depth seismicity—beyond 200 km from the trench and depths larger than 75 km. The importance of this is that earthquakes occurring in these sources have different potential hazard, evidenced by higher ground accelerations for intermediate depth earthquakes caused by higher stress drops or by source location, compared to coastal earthquakes, particularly demonstrated in Chile in the last 30 yr (Kausel & Campos 1992; Astroza *et al.* 2005; Leyton *et al.* 2009; Otarola *et al.* 2021).

Northern Chile is a seismically active region suitable for testing this hypothesis, having high seismicity rates and a seismological network that allows studying statistical properties of seismicity. We use the catalogue published by (Sippl *et al.* 2018a) that exploited the IPOC seismic network (GFZ German Research Centre For Geosciences & Institut Des Sciences De L'Univers-Centre National De La Recherche CNRS-INSU 2006) records, and also the USGS catalogue compiled by Poulos *et al.* (2019). By first formulating the unified scaling law and, then evaluating it on the catalogues we show that: (i) the scaling function is not universal and depends on whether the seismicity is located well within the crust or not; (ii) scaling exponents also differ and and (iii) waiting time probability distribution D in the long scales has power-law decay, implying long-term correlations.

We start in the next section formalizing the dimensional analysis approach to seismicity analysis, specifying hypotheses commonly assumed (Section 2). Then data and context are exposed in Section 3. Section 4 presents methodology used to evaluate scaling relations. Section 5 presents results, Section 6 discussion and Section 7 the conclusions.

2 SIMILARITY PROPERTIES OF SEISMICITY

2.1 Incomplete similarity and intermediate asymptotics in seismicity

Scaling can be seen as the procedure of relating values of a property from one scale to another (Hunt & Ewing 2016). A formalization of this procedure was proposed by Barenblatt (2003), who starts by writing a relation between the inquired property and a number of governing parameters. In a first instance, the relation can be expressed in dimensionless form using classical dimensional analysis and the Pi theorem (Buckingham 1914), thus reducing the number of relevant arguments. This theorem tells us that any physically significant relation (one that express a valid law to any observer, in particular those with unit measures of different magnitude) between dimensional parameters can be reduced to a relation between $p = n - k$ dimensionless parameters, with n the number of governing dimensional parameters and k the number of independent dimensions involved. Once dimension reduction is carried out then it can be determined what kind of similarity exists between the inquired property and its dimensionless arguments. If the property does not change under a change in the value of a certain dimensionless parameter, then we claim that the property displays complete similarity on that parameter and it can be ruled out from the formulation. On the contrary, a more general case is incomplete similarity where the dependence holds. In incomplete similarity a scaling function exists having as arguments powers of dimensionless parameters. Representations arising from similarity considerations are linked to the concept of intermediate asymptotics. The property, or more generally the phenomena under consideration, is analysed 'at intermediate times and distances away from boundaries such that the effects of accidental features or finer details in the spatial structure of the boundaries have disappeared but the system is still far away from its ultimately equilibrium state' (Barenblatt 2003).

2.2 Scaling of earthquake waiting time probability distribution

In the view of seismicity as a stochastic marked point process, in which the mark in the time-series corresponds to a scalar quantity and in this case to the earthquake seismic moment, the most important characteristic is waiting time τ. Here it is described using a seismic catalogue data with N events obtained in a observation time τ_0, with cut-off moment M_c, localized in a seismogenic volume of characteristic linear size L_0 and produced at a mean rate $\lambda_0 = N_0/\tau_0$. Plus, it is desirable to characterize D for different exceedance moments M, and different target epicentral areas of linear dimension L. Then, D corresponds to the probability density of waiting times τ from events having moment larger or equal than M, localized in the same epicentral area of linear dimension L, from a catalogue of events with moment equal or larger to M_c, localized in a seismogenic volume of linear size L_0 and produced at a rate λ_0:

$$D = f(\tau, M, L, \lambda_0, M_c, L_0), \tag{2}$$

where f is determined through data analysis. Using the unit system class TML (time, moment, length), M_c, L_0 and λ_0 are selected as parameters with independent dimensions. Dimensional analysis allows us to express D in dimensionless form $\Pi = D/\lambda_0$ and to propose three dimensionless parameters:

$$\Pi_\tau = \frac{\tau}{\lambda_0^{-1}}$$
$$\Pi_M = \frac{M}{M_c}$$
$$\Pi_L = \frac{L}{L_0}, \tag{3}$$

Then, the Pi theorem allows us to express our relation (eq. 2) in terms of a function Ψ of 3 instead of 6 parameters:

$$\Pi = \Phi(\Pi_\tau, \Pi_M, \Pi_L), \tag{4}$$

or in terms of the original variables:

$$D/\lambda_0 = \Phi(\lambda_0\tau, M/M_c, L/L_0). \tag{5}$$

Now we can elucidate the similarity of waiting time distribution with respect to these parameters. Let us introduce λ as the number of events per unit time that takes place in an epicentral area of linear dimension L with seismic moment larger than M. Classical statistical observations account for the scaling of λ as a power of L and M (Gutenberg & Richter 1965; Hanks & Kanamori 1979; Kagan & Knopoff 1980; Okubo & Aki 1987; Aviles *et al.* 1987; Kosobokov & Mazhkenov 1994): $\lambda(M, L) \propto M^{-\beta} L^{\gamma}$. We then hypothesize that the fraction of events per unit time that return after a waiting time τ in an epicentral area of linear dimension L with seismic moment larger than M also scales with powers of M and L. In our case, this corresponds to the scaling of D through its dimensionless parameters Π_M and Π_L, which can be formulated as a case of incomplete similarity:

$$\Phi = \Pi_M^{\beta} \Pi_L^{\gamma} \Psi \left(\frac{\Pi_\tau}{\Pi_M^{\beta_\tau} \Pi_L^{\gamma_\tau}} \right), \tag{6}$$

where β, γ, β_τ and γ_τ are exponents and Ψ is a function.

Additionally imposing $\beta_\tau = -\beta$; $\gamma_\tau = -\gamma$, we can write:

$$D = \lambda_0 \left(\frac{M}{M_c} \right)^{\beta} \left(\frac{L}{L_0} \right)^{\gamma} \Psi \left(\lambda_0 \left(\frac{M}{M_c} \right)^{\beta} \left(\frac{L}{L_0} \right)^{\gamma} \tau \right). \tag{7}$$

This is a key assumption and implies that exponents does not depend on the waiting time. Finally, replacing $\lambda = \lambda_0 \left(\frac{M}{M_c} \right)^{\beta} \left(\frac{L}{L_0} \right)^{\gamma}$ we have:

$$D = \lambda \Psi (\lambda \tau), \tag{8}$$

a expression that account for similarity of waiting time distribution on the seismic rate λ, generalized homogeneity of scaling function Ψ and incomplete similarity with seismic moment and epicentral area linear dimension. The quantity $\lambda\tau$ is called the renormalized waiting time because it is twice normalized, both by the mean rate and by the powers of M and L. For the same reasons, D/λ is called the renormalized waiting time density.

The scaling function Ψ determines how the probability of observing a recurring event changes with waiting time τ elapsed from the last one, the seismic moment (or magnitude) exceedance M and the epicentral area linear dimension L, through the renormalized waiting time $\lambda\tau$. Christensen *et al.* (2002) analysis of California seismicity shows a regime coherent with a $\Psi \propto \tau^{-1}$ power-law scaling function in the short $\lambda\tau$-scale, later corroborated theoretically by Saichev & Sornette (2007). A transition was found around $\lambda\tau = 1$, that is, event-pairs with waiting time close to the mean waiting time in their own L and M scale. Then a more pronounced decay of Ψ is observed for larger $\lambda\tau$-scale. This regime was first described in terms of a Poisson process for main shocks, but later in terms of a power law (Corral 2003).

3 REGIONAL SETTING AND DATA

In northern Chile, as shown in Fig. 1, there is a convergent contact, where the Nazca Plate advances at 68 mm a^{-1} (Norabuena *et al.* 1998) in the direction N76 E (Angermann *et al.* 1999) with respect to the South American continent. The trace of this convergence (trench) is located roughly in a north–south direction at the greater bathymetric depths. Under the continent the subducting plate is localized, showing a simple but abrupt morphology (Contreras-Reyes *et al.* 2012). From 2001 onwards destructive earthquakes of different types have been recorded, with the exception of outer-rise events: at the plate interface the 2014 M_w 8.1 Iquique earthquake (Ruiz *et al.* 2014) and the 2007 M_w 7.7 Tocopilla earthquake (Delouis *et al.* 2009), the intermediate-depth 2005 M_w 7.8 Tarapacá earthquake (Peyrat *et al.* 2006) and the crustal 2001 M_w 6.3 Aroma earthquake (Legrand *et al.* 2007).

Fig. 1 also shows the modern station network with spatial and azimuthal coverage that has allowed the computation of high-resolution seismic catalogues. We use mainly the catalogue published by Sippl *et al.* (2018b, a) which makes use of records acquired by the Chilean National Seismological Center (see Barrientos & National Seismological Center (CSN) Team 2018) and IPOC networks, in addition to other temporary stations between the years 2007–2014. This catalogue contains 101602 events relocated with the double-difference method (Waldhauser & Ellsworth 2000). These events are distributed between 0.01 and 270 km depth, with location error varying between 1.5 and 15 km depending on hypocentre location. Reported magnitudes range from 1.3 to 8.1, although according to Hainzl *et al.* (2019) the catalogue is complete for magnitudes $M_\mathrm{l} \geq 2.7$. For comparison sake we also use USGS catalogue for years 1974–2017 compiled and homogenized to moment magnitude (M_w) by Poulos *et al.* (2019). This catalogue has 863 events in Northern Chile with completeness magnitude of 5.0. Fig. 1 shows the spatial distribution of the seismic events. Two seismicity groups can be observed according to longitude: seismicity distributed around coastal zone and around the volcanic arc. Coastal seismicity is mainly associated with the subduction interface between plates. Intermediate depth seismicity corresponds to earthquakes between 80 and 270 km depth.

These two groups can be observed in the cross-section presented in Fig. 2, where the distinction between coastal and intermediate depth seismicity can be clearly seen. The former has a spatial distribution of hypocentres along a dipping plate, with clusters of seismicity separated by some vertical offset, and upper plate seismicity above this dipping plate configuration. Intermediate depth seismicity is composed by a cluster of hypocentres that fills gaps present between thin plates in coastal seismicity (Sippl *et al.* 2019). Both groups are separated by a minimum in the number of earthquakes around 75 km depth, as shown in the density plot on the right-hand side of Fig. 2, visible also in the USGS catalogue. This rough separation in coastal seismicity with depth $<$75 km and intermediate depth seismicity with depth $\geq$75 km will be analysed in subsequent sections. Justification for the occurrence of this two distinct groups can be found on literature, Oncken *et al.* (2003) for example showed that subducted Nazca Plate reflector is no longer observed at 80–90 km due to the 650–700° isotherm that prevents stability of serpentinized oceanic crust.

To further inquire about differences between seismicity groups we will also analyse cluster distribution proposed by Sippl *et al.* (2018b) based on a visual inspection of hypocentre geometry, according to criteria in table 1 of their paper. Coastal seismicity was subdivided into three clusters: a well-defined double seismic zone consisting of the upper P2 plane and the lower P3 plane, an observation previously noted by Comte *et al.* (1999), Dorbath *et al.* (2008), Rietbrock & Waldhauser (2004), Bloch *et al.* (2014) and Brudzinski *et al.* (2007); P1 interface seismicity above the P2 plane associated with thrust-type earthquakes (Delouis *et al.* 1996). Crustal seismicity is labelled as UP (upper plate), and although is most abundant above plate interface, it is also present around the Andes cordillera (Herrera *et al.* 2021). Events at depths larger than 180 km were not considered due to poor station coverage.

4 METHODS

Empirically evaluating the scaling relationship (eq. 7) requires analysing the temporal distribution of seismicity in the study region for different seismic moment ranges and different epicentral area linear dimension ranges. As the IPOC catalogue contains only local magnitudes, ratios between two seismic moments M_1 and M_2 will be of the form $M_1/M_2 = 10^{\frac{3}{2}d(Ml_1 - Ml_2)}$, where Ml_1 and Ml_2 are the respective local magnitudes and d is the slope in the moment-magnitude-local-magnitude linear regression. As in this case d is unknown, we will assume $d = 1$, which will not influence the consistency of results, but could influence consistency in comparing

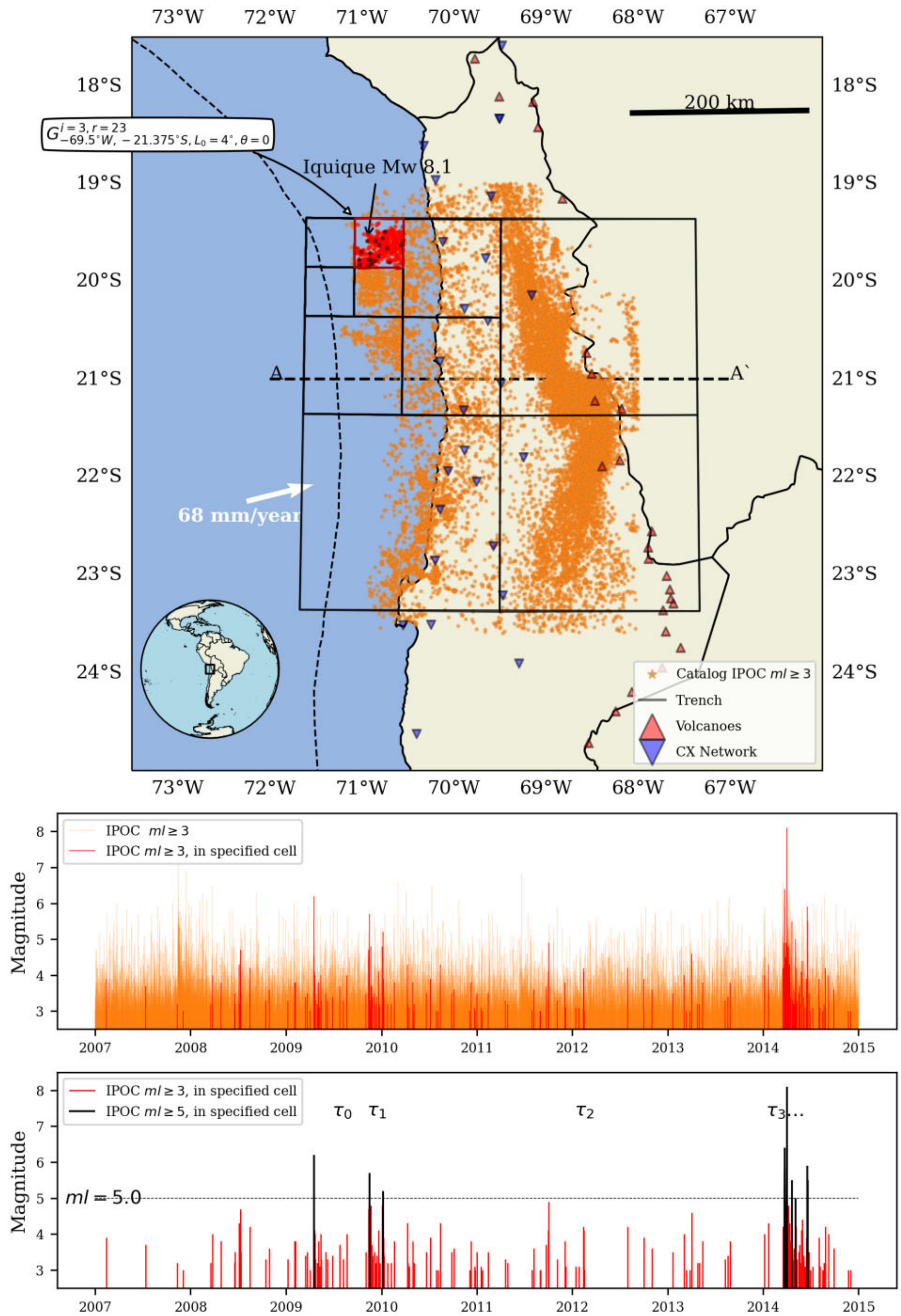

Figure 1. Top: Study zone in northern Chile between 67–73°W and 18–24°S. Permanent stations from IPOC CX Network (blue inverted triangles), active volcanoes (red triangles) and trench trace (segmented black line) are shown. White arrow shows the direction and magnitude of plate convergence vector. Epicentres from IPOC catalogue with $M_l \geq 3$ are plotted (orange points). A hierarchic grid is positioned centred at 69.5°W, 21.375°E with base level linear dimension 4° (with 1° = 111 km), so that epicentres are well enough covered. A specific grid cell from hierarchy level $i = 3$ is indicated, with epicentres plotted in red. The subcatalogue generated through selection of catalogue events from a given cell is shown in time-magnitude in the central panel. Lower panel shows $M_l \geq 5$ magnitude slice filtering also plotted as black points in the map. From this type of subcatalogues waiting times τ can be obtained. Black arrow show the M_w 8.1 Iquique earthquake epicentre. A–A' cross-section is shown in Fig. 2.

scaling coefficients with other seismic catalogues. Then, seismic moment ranges for IPOC catalogue are defined from exceedance magnitudes for $Ml = 3.0$, 3.5, 4.0 and 4.5 slices. It is important to note that magnitude 3.0 is arbitrary, but it must be larger than the catalogue cut-off magnitude. For USGS catalogue in turn, seismic moment ranges are defined from exceedance magnitudes for $M_w = 5.0$, 5.5 and 6.0 slices.

Epicentral area linear dimension ranges are defined from hierarchical grids with base level size $L_0 = 4°$ which covers the whole seismogenic area, with $1° = 111$ km. UTM geographic projection is used. Each grid is constructed by progressive subdivisions $L_i = L_{i-1}/4$, with $i = 1, 2, 3$ and 4, with 4,16,64 and 128 cells of size $L_1 = 2°$, $L_2 = 1°$, $L_3 = 0.5°$ and $L_4 = 0.25°$, respectively. The grid used for the calculations is shown in Fig. 1.

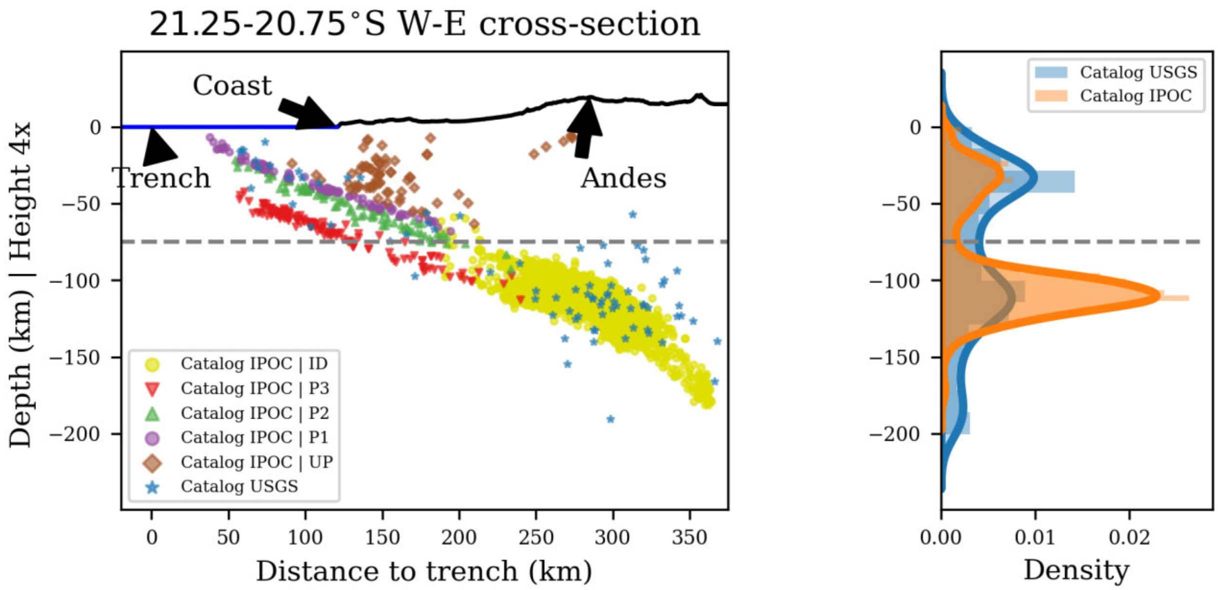

Figure 2. Left: Cross-section A–A' at 21°S between 72 and 67°W as shown in Fig. 1. Abscissa represents distance to trench in km, ordinate represents depth with respect to sea level in km. Topography is exaggerated four times. Features indicated are trench, coastline and Andes. Seismicity with $5.0 \leq M_{\mathrm{w}}$ from USGS catalogue (with crustal events discarded) and with $3.0 \leq M_{\mathrm{l}}$ for IPOC catalogue is shown for events located between 21.25 and 21.75°S. In colours, according to the legend, are indicated the different seismicity clusters proposed by Sippl *et al.* (2018b): ID, Intermediate-Depth Cluster; P3, Lower Plane; P2, Upper Plane; P1, Plate Interface and UP, Upper Plate Cluster. Right: hypocentre depth density distribution for the IPOC catalogue and also for the 1974–2018 USGS catalogue, both having two distinct modes: coastal and intermediate depth seismicity, separated by a minimum at ∼75 km depth, indicated by the grey discontinuous line.

With the previously defined M and L ranges it is possible to proceed in obtaining samples of waiting times. This is achieved through subcatalogue generating process, by assigning cell coordinates to the whole catalogue using a quadtree search and filtering with moment ranges defined by slices. This process is illustrated in Fig. 1. Waiting times samples obtained from different cells are treated equally, this is a hypothesis of seismic homogeneity and isotropy, that is D waiting time probability distribution does not depend on the cell location, nor on cell orientation. In this way it is possible to say that the obtained empirical distribution comes from a cell chosen at random, which allows to collect observations from all cells in one set.

Apart from obtaining waiting time samples, mean annual number of events N_{ji} for each spatial level i and for each seismic moment j are also obtained. This allows to estimate β and γ coefficients according to the method proposed by Kosobokov & Mazhkenov (1994) (SCE, Scaling Coefficient Estimation) and suggested as a proper renormalization factor for the waiting time distribution by Molchan (2020). Details regarding coefficients estimation procedure can be found on Nekrasova & Kossobokov (2020). It involves least-squares inversion from linear equations:

$$\log N_{ji} = \Lambda - \beta(M_j - M_c) + \gamma \log(L_i/L_0), \quad (9)$$

where Λ represents the estimation of the mean seismic rate, as the logarithm of the annual number of earthquakes with seismic moment equal or larger than M_c in an area of linear dimension L_0. Finally, to stabilize the coefficient estimation and obtain confidence intervals we repeat the sampling process by performing 100 random rotations of the grid around its centre.

Once waiting times are renormalized, we provide graphical visualizations on log-log plots. Please note that both axes are a dimensional, so the axis scale is set to 1:1 in order to adequately infer qualitative characteristics. Previous applications do not provide equally scaled axes, which can distort the scaling function.

Next, we will assess the presence or absence of the power-law asymptotic for short and long scales using least squares and the likelihood method suggested by Clauset *et al.* (2009), respectively. The short-scale asymptotic is described qualitatively, and the long scale requires an assessment of its heavy-tailedness, for which Clauset *et al.* (2009) log-likelihood ratio between power-law and exponential fit test is most appropriate.

Additionally, a graphical comparison will be conducted with an exponential decay of waiting times $\Psi(\lambda\tau) = \exp(-\lambda\tau)$, which is the scaling function that models waiting times in a homogeneous Poisson process (Langenbruch *et al.* 2011).

5 RESULTS

Table 1 shows the scaling coefficients obtained through least-squares inversion of the estimated renormalization factors. Λ corresponds to the logarithm of the estimated annual number of earthquakes taking place in a cell of linear dimension L_0 with seismic moment larger or equal than M_c. Then, Λ essentially points to seismic activity intensity; greater in intermediate depth clusters and interface seismicity.

β corresponds to the balance between number of earthquakes generated by magnitude range; this coefficient is smaller in the case of interface seismicity, attributable to the presence of mega-thrust earthquakes. Lower b-values for thrust events have been previously reported in the literature (e.g. Schorlemmer *et al.* 2005; Petruccelli *et al.* 2019). It is notable that it increases with depth from slab, this is in accordance with b-values $(\frac{3}{2}\beta)$ obtained by Hainzl *et al.* (2019).

Table 1. Scaling coefficients Λ, β, γ and least-square residuals (RES) associated to renormalization factor $\hat{\lambda}$ estimation. Values shown as the median of distribution with 95 (up) and 5 (down) percentile.

Catalogue	Λ	β	γ	RES
USGS $z < 75$ km	$0.85^{0.05}_{0.10}$	$0.59^{0.05}_{0.04}$	$1.16^{0.09}_{0.08}$	$0.04^{0.04}_{0.02}$
USGS $z \geq 75$ km	$0.97^{0.02}_{0.08}$	$0.82^{0.03}_{0.07}$	$1.32^{0.04}_{0.10}$	$0.06^{0.11}_{0.02}$
USGS all	$1.11^{0.03}_{0.05}$	$0.65^{0.04}_{0.04}$	$1.45^{0.06}_{0.07}$	$0.09^{0.05}_{0.03}$
IPOC $z < 75$ km	$2.74^{0.06}_{0.08}$	$0.45^{0.01}_{0.01}$	$1.23^{0.05}_{0.06}$	$0.06^{0.04}_{0.03}$
IPOC $z \geq 75$ km	$3.33^{0.02}_{0.04}$	$0.64^{0.01}_{0.01}$	$1.22^{0.02}_{0.02}$	$0.04^{0.02}_{0.01}$
IPOC all	$3.40^{0.01}_{0.01}$	$0.61^{0.01}_{0.02}$	$1.40^{0.04}_{0.04}$	$0.06^{0.03}_{0.03}$
IPOC UP	$1.91^{0.02}_{0.05}$	$0.58^{0.02}_{0.02}$	$1.22^{0.14}_{0.14}$	$0.03^{0.06}_{0.01}$
IPOC P1	$2.50^{0.04}_{0.07}$	$0.43^{0.02}_{0.02}$	$1.11^{0.06}_{0.07}$	$0.07^{0.07}_{0.04}$
IPOC P2	$2.19^{0.05}_{0.11}$	$0.50^{0.02}_{0.02}$	$1.27^{0.06}_{0.10}$	$0.05^{0.04}_{0.02}$
IPOC P3	$1.80^{0.02}_{0.04}$	$0.52^{0.05}_{0.03}$	$1.30^{0.02}_{0.04}$	$0.03^{0.02}_{0.02}$
IPOC ID	$3.32^{0.01}_{0.04}$	$0.64^{0.01}_{0.01}$	$1.20^{0.02}_{0.02}$	$0.04^{0.03}_{0.01}$

Lastly, γ corresponds to the correlation fractal dimension. It is the closer to 1 in the case of interface seismicity (1.1) and increases through upper (1.27) and lower (1.3) planes; with the UP and ID clusters having intermediate value $\sim$1.2. Same observation for larger fractal dimension with depth is valid for USGS catalogue, although not in IPOC for this basic 75 km separation. Residuals are in the median lesser than 0.1, but considerably larger in the cases of coastal seismicity and P1.

Fig. 3 shows the renormalized waiting time distribution as a function of the renormalized waiting times for the two catalogues (circumflex diacritic on quantities denotes empirical estimates). An overall collapse of the distributions resulting from the renormalization process is observed. It can be noted that for $\hat{\lambda}\tau < 10^{-2}$ divergence of curves is observed, not collapsing all in one function, most notably for the IPOC catalogue. This can be due to the undersampling of very short waiting times and/or due to the mixing of distributions from different scaling and functions and/or coefficients. This latter factor was noted by Davidsen & Goltz (2004), they found that different values of fractal dimensions where able to fit different regions of waiting time distribution in Southern California and Iceland catalogues. Above the curves we plotted the empirical joint density function of all sampled renormalized waiting times (rather than separated in their respective *M* and *L* ranges), which is marked by the curve (blue) corresponding to the distribution obtained by considering all interevent times (indicated as $\hat{D}_{\hat{\lambda}T}$). This curve represents the overall behaviour of scaling function Ψ, although it can be noted that for $\hat{\lambda}\tau < 10^{-2}$ it tends to go below individual distributions, mostly.

Scaling function Ψ is generally decreasing for all clusters. It can be noted that the ordinate value $\hat{D}\hat{\lambda}^{-1} = 1$ marks the transition from a gentler to a stepper descent, with a middle slope in between that ends around the abscissa $\hat{\lambda}\tau = 1$. This means points located to the left of the transition correspond to recurrences occurring with frequency larger than expected from the mean rate: $\hat{D} > \hat{\lambda}$; and vice versa. Moreover, renormalized waiting times with $\hat{\lambda}\tau < 1$ means recurrences having waiting times shorter than the mean waiting time $\hat{\lambda}^{-1}$; and vice versa. Then, transition region is composed by recurrences with waiting times shorter than the mean waiting time that occur with less frequency than the expected from the mean rate. This transition region will be important when comparing waiting time distributions for different groups and cluster, as shown next. Fig. 4 allows us to compare scaling functions Ψ for the USGS and IPOC catalogues in the coastal ($z < 75$ km) and intermediate depth seismicity groups ($z \geq 75$ km). Furthermore, analysis for clusters UP, P1, P2, P3 and ID are shown in Fig. 5. Both figures are organized such as to easily visualize the differences. Left-hand columns with coastal seismicity and clusters UP, P1 and P2 posses a steep power-law descent in the region before the transition, with coefficients $\alpha \sim 0.7$ in the case of UP and P2 and near 0.9–1 in other cases, comparable to Omori coefficient for aftershocks. Right-hand columns in turn, with intermediate depth seismicity, P3 and ID clusters show gentler slope power-law segments with $0.2 < \alpha < 0.6$. The other remarkable difference between columns is the form and values of the Ψ scaling function in the transition region. For comparison, a exponential function is shown in each figure. In the left columns coastal seismicity, UP, P1 and P2 clusters scaling functions do not agree with a exponential decay in the transition region. In fact, $\Psi < \exp(-\hat{\lambda}\tau)$, meaning that observed recurrence density is less abundant than the expected for a pure Poisson process. In turn, in the right columns intermediate depth seismicity, ID and P3 clusters have scaling functions Ψ that agree well with a exponential function in the transition region.

Finally, after the transition region all scaling functions seem to posses a power-law decay. We tested this assumption with likelihood ratio between power-law and exponential fits (Clauset *et al.* 2009), obtaining positive values in all cases, but significance only for IPOC catalogue except UP and P3, see Table A1 in the Appendix, which could be attributed to scarcity of data in those clusters and in USGS catalogue.

6 DISCUSSION

We discuss Fig. 6, in which the seismicity will be analysed for the following groups:

(i) Upper Seismicity Group, including seismicity above 75 km depth and individual UP (crustal), P1 (interface) and P2 (upper plane from double seismic zone) clusters.

(ii) Lower Seismicity Group: including seismicity below 75 km depth and individual ID (intermediate depth) and P3 (lower plane from double seismic zone) clusters.

Region A corresponds to clustered seismicity, it is associated with aftershocks, foreshocks and swarms. This region is composed by recurrences with waiting times that are smaller than

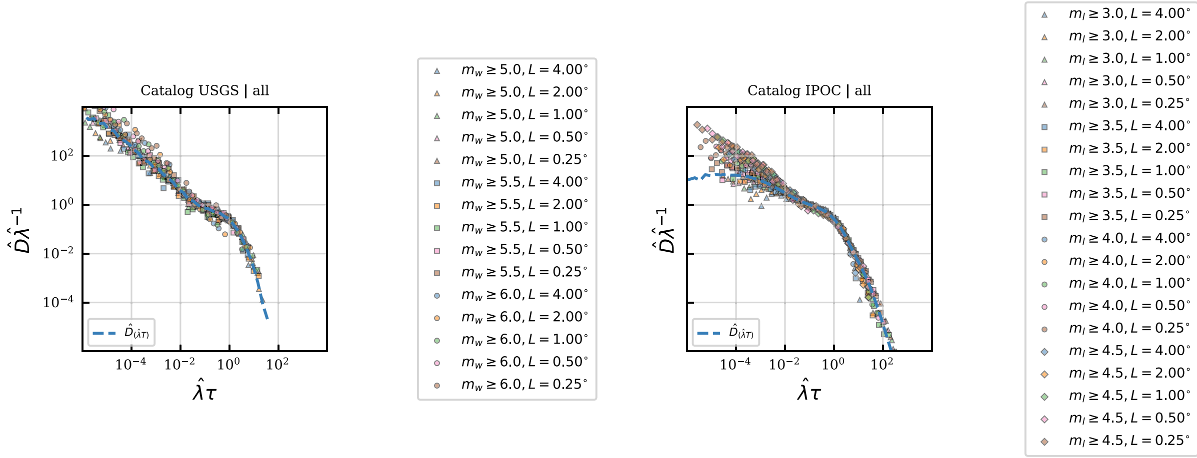

Figure 3. Renormalized empirical waiting time density $\hat{D}\hat{\lambda}^{-1}$ (ordinate) versus renormalized waiting time $\hat{\lambda}\tau$ (abscissa), for USGS (left-hand panel) and IPOC (right-hand panel) catalogues. Coloured markers indicating different magnitude and epicentral area linear dimension ranges as shown in the legend. Blue curve corresponds to joint empirical density $\hat{D}_{(\hat{\lambda}T)}$.

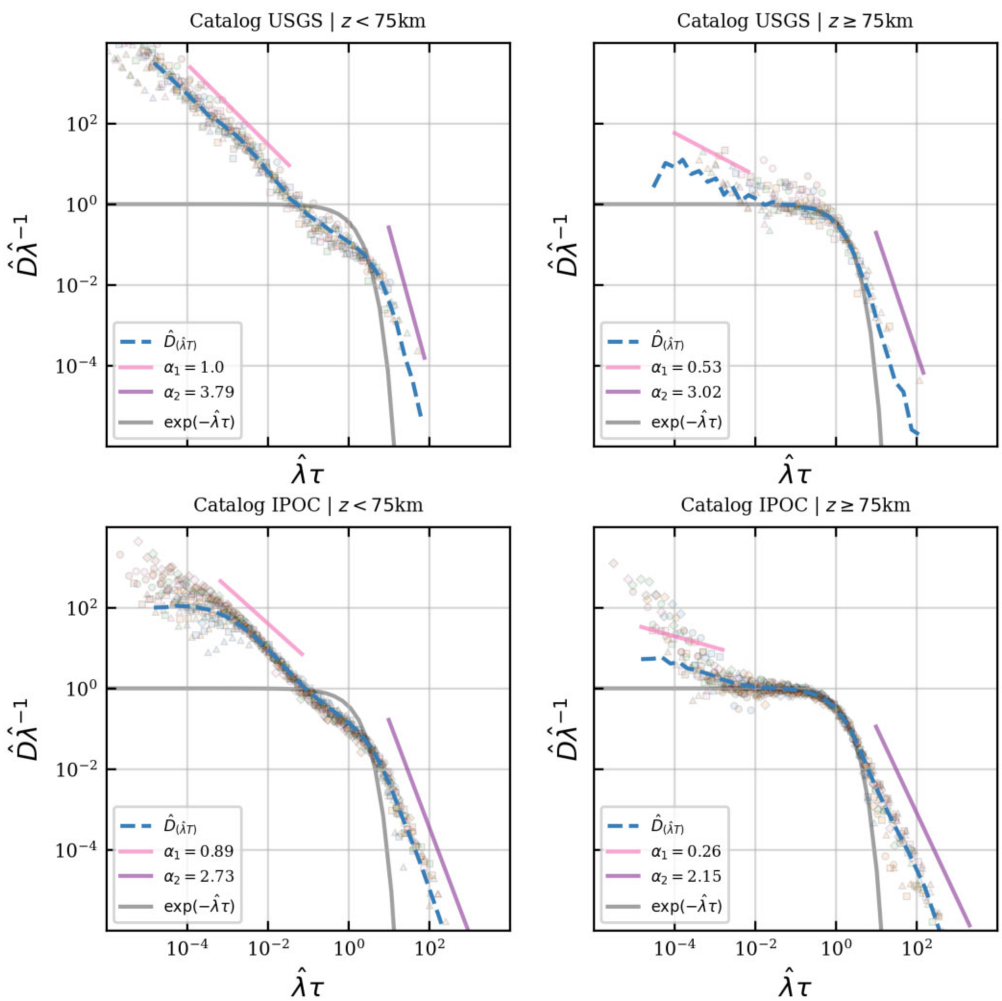

Figure 4. Renormalized empirical waiting time density $\hat{D}\hat{\lambda}^{-1}$ (ordinate) versus renormalized waiting time $\hat{\lambda}\tau$ (abscissa) for USGS and IPOC coastal (left-hand panel) and USGS and IPOC intermediate depth (right-hand panel). Dashed blue curve corresponds to joint empirical density $\hat{D}_{(\hat{\lambda}T)}$. Solid lines corresponds to fitted power-law segments and exponential scaling function. Markers are the same as for previous figures, representing individual scale distribution.

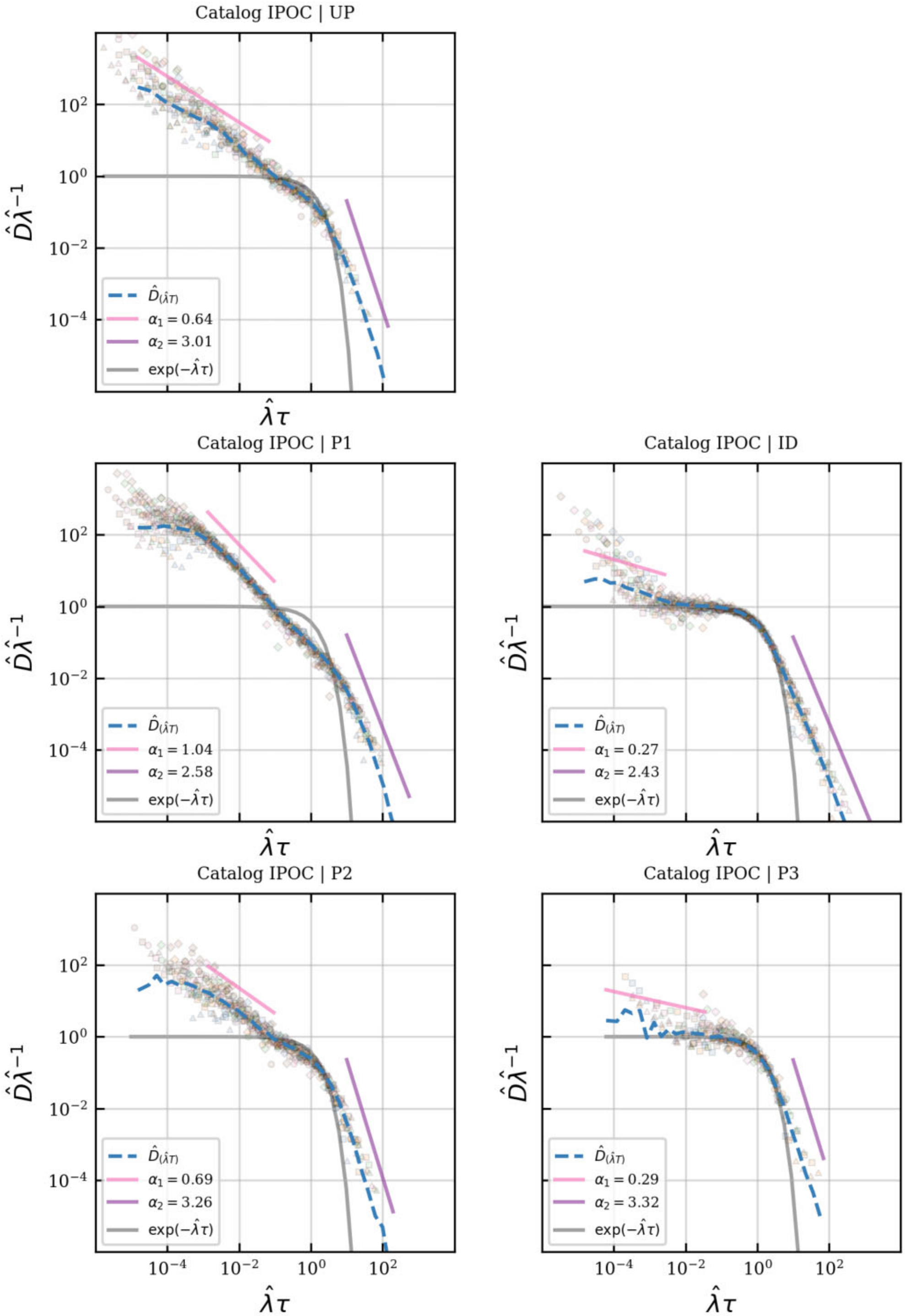

Figure 5. Renormalized empirical waiting time density $\hat{D}\hat{\lambda}^{-1}$ (ordinate) versus renormalized waiting time $\hat{\lambda}\tau$ (abscissa) for IPOC UP, P1 and P2 clusters (left-hand panel) and IPOC intermediate depth P3 and ID clusters (right-hand panel). Dashed blue curve corresponds to joint empirical density $\hat{D}_{(\hat{\lambda}T)}$. Solid lines corresponds to fitted power-law segments and exponential scaling function. Markers are the same as for previous figures, representing individual scale distribution.

the mean waiting time ($\tau < \hat{\lambda}$) for which the observed density is larger than the mean rate ($\hat{D} > \hat{\lambda}$). Besides, a probability surplus with respect to a exponential function is observed. The fitted power-law scaling exponent α_1 modulates the degree of productivity, stronger for the Upper Seismicity Group and weaker for the Lower Seismicity Group. The limiting value $\alpha_1 = 1$, close to the one fitted for interface cluster—and coastal seismicity, in general—is consistent with Omori Law (e.g. Saichev & Sornette 2007).

Region B corresponds to recurrences which occur after waiting times that are close to the mean waiting time $\tau = \hat{\lambda}^{-1}$. Also, from it's upper limit and downwards the observed density of recur-

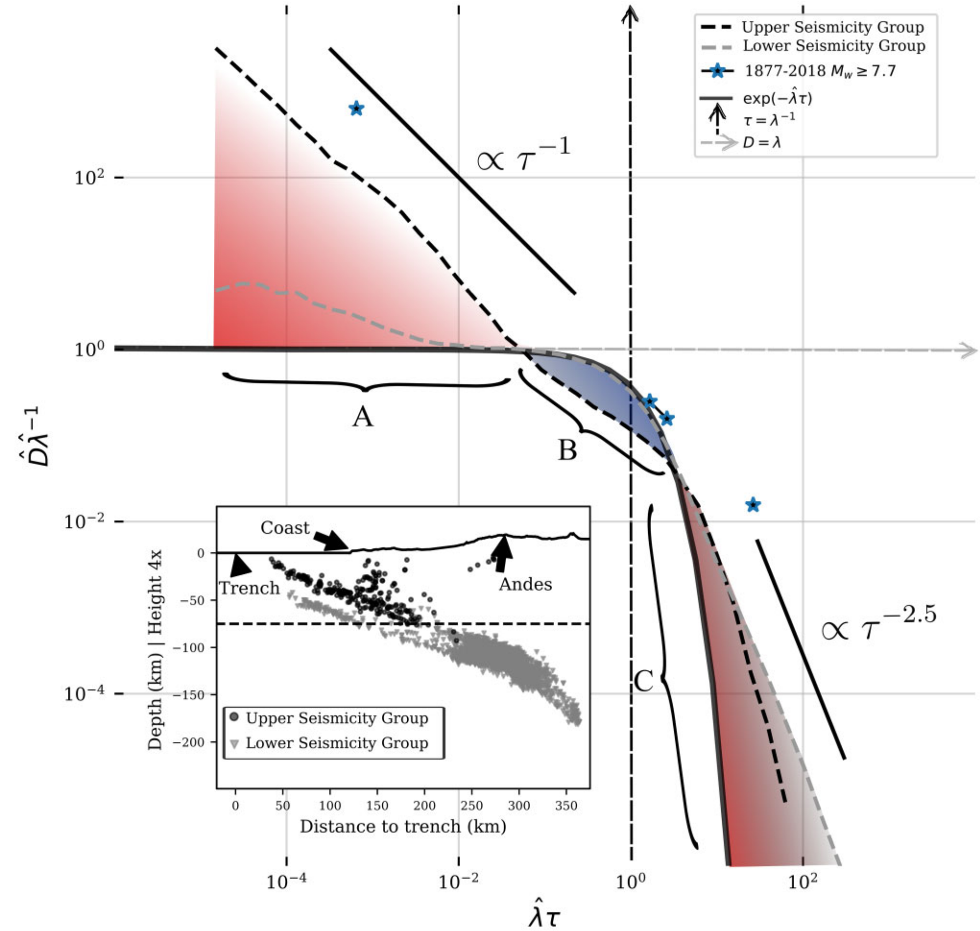

Figure 6. A sketch of the earthquake waiting time distributions trough its scaling function and its three distinctive A, B and C regions; for Upper Seismicity Group and Lower Seismicity Group, which are shown in the lower left corner inset in a thin section. It is also shown the scaled waiting time distribution of 1877–2018 $M_w \geq 7.7$ thrust earthquakes with star markers.

rences is less than expected from the mean rate ($\hat{D} < \hat{\lambda}$). In this region, a probability deficit with respect to exponential function is appreciated for the Upper Seismicity Group. The Lower Seismicity Group, on the other side, agrees well with the exponential function. This latter observation was also made for intermediate depth seismicity in Vrancea region (Enescu *et al.* 2008)and for fluid induced seismicity (Langenbruch *et al.* 2011). Observations also agree well with those of Telesca *et al.* (2020) performed on the same IPOC catalogue, they found interface seismicity clustering persists even after declustering the catalogue, thus correlations dominate, while lower plane seismicity maintains its Poisonnean nature.

Region C corresponds to recurrences with waiting times much longer than the mean waiting time. In this region all divisions have power-law decay of their scaling functions. This means a surplus of probability with respect to the exponential function. Power-law behaviour is related to long-term clustering (e.g. Kagan & Jackson 1991). Recent papers on global long-term scale agree with departure from exponential decay (Griffin *et al.* 2020; Chen *et al.* 2020). Although the statistical test to verify power-law does not yield positive results in medium-term USGS catalogue, it is clear from results that scaling functions posses a decay that is slower than a exponential; this could be modelled with Gamma, Weibull or other distributions (e.g. Corral 2003; Abaimov *et al.* 2008; Mangira *et al.* 2019).

What are the consequences for the expected time to the next event? If for a specified moment-area scale (i.e. defining M corresponding to a certain magnitude threshold and L) an earthquake has been waited for a time short enough (according to scale), then it can be expected, with probability larger than the one obtained by assuming events to be randomly distributed, that another one will occur close in time. This probability is larger, in general, for near coast epicentres, and particularly for the Upper Seismicity Group, once its hypocentre is finely estimated. Please note that as no distinction is made between foreshocks, main shocks and aftershocks, every other earthquake taking place restarts the waiting time count. Thus, if an earthquake has been waited sufficiently long to enter region B, it is expected that it occurs with a probability lesser than what a random distribution of waiting times shows, in the case of the Upper Seismicity Group, and with a probability just like it, for the Lower Seismicity Group. Finally, if waiting time count continues past the transition and enters region C, the probability is larger than what a random-distributed model of waiting times shows.

How does historical, large $M_w \geq 7.7$ thrust earthquakes look in this picture? In Fig. 6 it is shown their scaled waiting time distribution in the same epicentral area with $L = 4^\circ$ shown in Fig. 1, that were obtained from Ruiz & Madariaga (2018) for years 1877–2014. It is possible to observe a point above region A, corresponding to short-time clustered large events (the time distance between Iquique M_w 8.1 main shock and its largest M_w 7.7 aftershock). Note that not only large 'aftershocks' with lower magnitude than a certain large event can develop closely in time, but also very large 'doublets' (Kagan & Jackson 1999) or 'foreshocks', as was the case with the farther south 1960 Valdivia seismic sequence (Ojeda *et al.* 2020). Then, there are two points above region B, corresponding to events close to the mean waiting time estimated from extrapolation of seismic rate for $z < 75$ km in USGS catalogue (the time distance between Iquique and Tocopilla M_w 7.7 earthquake and the time distance between Tocopilla and 1995 M_w 8.5 Antofagasta earthquake), and one point in region C, corresponding to the time distance between large tsunamigenic 1877 earthquake and 1995 Antofagasta earthquake). In all cases, it can be noted that density is larger than the overall tendency represented by scaling function Ψ. Caution in the interpretation of this is needed, due to the short time span of the catalogue used in comparison to the waiting time between major earthquakes.

We may discuss a simple tectonic interpretation of the differences observed. Upper Seismicity Group corresponds to a crust-on-crust system, while the Lower Seismicity Group corresponds to a crust–mantle system (e.g. The ANCORP Working Group 1999; Peacock 2001; Sippl *et al.* 2019). The crust-on-crust system posses a different topology—the one conveyed by a physical interface—and a larger dimensionality, capable of higher interconnectedness between faults. Interactions, thus produce larger correlations between events, shown as sharp power-law decay in the short scale, departure from exponential function in the central transition regime and long-term clustering. This is in accordance with the key role played by the volumetric abundance of nearby faults (Dascher-Cousineau *et al.* 2020), which at the same time is capable of explaining why in Northern Chile aftershock productivity decreases with decreasing coupling (Hainzl *et al.* 2019) and with depth from slab (Cabrera *et al.* 2021).

Interestingly enough, crustal seismicity also posses the kind of behaviour described for interface seismicity, which is in accordance with results obtained by Stallone & Marzocchi (2019). Although rather infrequent compared to large interface events, crustal earthquakes pose a major risk to highly populated cities (e.g. Yang & Yao 2021; Cui *et al.* 2011; Suárez *et al.* 2019; Pérez *et al.* 2014).

Is there any relation between space-moment scaling coefficients β and γ and the described waiting time distributions? Our results indicate that interface seismicity have lower β hence *b*-value and fractal dimension than other clusters. This could be due to interface seismicity in northern Chile occurring in a somehow smooth seismogenic contact, as recently suggested from subducted seafloor analysis (Lallemand *et al.* 2018) or from theoretical formulation (Venegas-Aravena & Cordaro 2023). Higher values of fractal dimension means epicentres filling a plane, while values closer to 1 means epicentres concentrated along linear features. In the case of interface seismicity, a smoother interface could mean localized stresses, being aligned with the stress field allowing earthquakes to grow big, hence reducing the *b*-value. This has been suggested from laboratory experiments (e.g. Goebel *et al.* 2017; Fryer *et al.* 2022). Besides, residual stresses could produce higher numbers of clustered seismicity and aftershocks, as shown in relationship between Omori coefficient and *b*-value from recent physical-based and data analyses results (Zaccagnino *et al.* 2022). In other clusters seismicity is clearly dispersed and more homogeneously distributed, suggesting a more fractured media hence raising fractal dimension, raising *b*-value and lowering short-scale clustering. We here need to call for caution since distributions for scaling coefficients from random sampling are skewed and sometimes bimodal, this could be due to latitudinal or temporal variations and more detailed studies and exploration in other subduction regions are needed in order to establish the significance of differences found and avoid biases (Marzocchi *et al.* 2020).

Due to results obtained in this study, we speculate that outer rise and deep focus earthquakes also posses distinct waiting time scaling behaviour, that future studies need to analyse. Furthermore, interactions between interface and other contexts is starting to be quantified [e.g. with intermediate depth Jara *et al.* 2017; Aden-Antóniow *et al.* 2020; Wimpenny *et al.* 2023, with crustal seismicity Pastén-Araya *et al.* (2021)]. Our scaling analysis can be considered minimalist since it contains only governing parameters obtainable from seismic catalogues. Nonetheless, it has the potential to incorporate more variables that could help to elucidate the origin of differences found in this study, detect interactions and allow the integration of seismicity catalogues with other kinds of data such as strain rates like those used in global seismicity models (e.g. Bayona *et al.* 2023), stress drops (Folesky *et al.* 2021) or radiated energy (Derode & Campos 2019).

7 CONCLUSIONS

In this study, we have used an approach based on a unified scaling law for seismicity, that we derived from similarity principles, to characterize the earthquake waiting time distributions from two seismic catalogues available for the Northern Chile subduction zone. Although the scaling approach only incorporate variables available in seismic catalogues, it is enough to provide evidence of distinct behaviours of seismicity for different spatial domains, as hypothesized. A rough division distinguishes coastal from intermediate depth seismicity, the former with marked spatio-temporal clustering behaviour and the latter with events more broadly spread in space and time. The availability of the IPOC catalogue with large spatial resolution and small magnitude completeness allowed us to capture an important distinction inside coastal seismicity: lower plane from the double seismic zone behaves like the intermediate depth seismicity. Furthermore, crustal upper plate seismicity and the upper plane from the double seismic zone appears with noticeable clustering as interface seismicity does. These observations permits us to provisionally classify the subduction system in 2 systems: seismicity in a crust-on-crust system and seismicity in a crust–mantle system. Our results are consistent with recent findings regarding the varying aftershock productivity and scaling parameters, providing a constraint to asses the generative earthquake mechanisms that governs a subduction zone. As stated in the introduction, earthquakes from these different contexts have varying potential hazard due to different source characteristics and ground acceleration. The classification of earthquakes according to spatio-temporal-moment organization provides an opportunity to enhance seismic hazard estimation and dissemination, which requires continuous efforts in enhancing seismological monitoring.

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DATA AVAILABILITY

The processed data and code used for making tables and figures in this study are available at Repositorio de Datos de Investigación de la Universidad de Chile via https://doi.org/10.34691/FK2/GGHMFZ.

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APPENDIX

Table A1. Power-law coefficients for short (α_1) and long (α_2) scale decay of scaling function, with respective standard deviations (σ_1 and σ_2). Values shown as the median of the distribution with 95 (up) and 5 (down) percentile. Likelihood ratio power-law test results PL test and associated p significance value are shown.

Catalogue	α_1	σ_1	α_2	σ_2	p	PL test
USGS $z < 75$ km	$-1.01^{0.05}_{0.03}$	$0.03^{0.01}_{0.01}$	$3.82^{0.56}_{0.87}$	$0.48^{0.19}_{0.16}$	0.47	False
USGS $z \geq 75$ km	$-0.52^{0.17}_{0.21}$	$0.11^{0.09}_{0.05}$	$3.04^{0.56}_{0.49}$	$0.54^{0.18}_{0.15}$	0.30	False
USGS Full	$-0.91^{0.06}_{0.04}$	$0.03^{0.01}_{0.01}$	$3.78^{0.57}_{0.56}$	$0.48^{0.11}_{0.10}$	0.51	False
IPOC $z < 75$ km	$-0.89^{0.02}_{0.03}$	$0.04^{0.00}_{0.00}$	$2.75^{0.19}_{0.24}$	$0.05^{0.01}_{0.01}$	0.00	True
IPOC $z \geq 75$ km	$-0.26^{0.04}_{0.06}$	$0.04^{0.02}_{0.01}$	$2.15^{0.09}_{0.15}$	$0.02^{0.00}_{0.00}$	0.00	True
IPOC Full	$-0.51^{0.04}_{0.03}$	$0.01^{0.00}_{0.00}$	$2.47^{0.09}_{0.06}$	$0.02^{0.00}_{0.00}$	0.00	True
IPOC UP	$-0.64^{0.02}_{0.02}$	$0.02^{0.01}_{0.00}$	$2.98^{0.53}_{0.16}$	$0.21^{0.08}_{0.03}$	0.17	False
IPOC P1	$-1.03^{0.03}_{0.02}$	$0.03^{0.01}_{0.00}$	$2.63^{0.09}_{0.16}$	$0.06^{0.01}_{0.01}$	0.03	True
IPOC P2	$-0.69^{0.03}_{0.03}$	$0.02^{0.01}_{0.01}$	$3.26^{0.38}_{0.24}$	$0.17^{0.02}_{0.03}$	0.01	True
IPOC P3	$-0.29^{0.17}_{0.14}$	$0.08^{0.05}_{0.03}$	$3.28^{0.33}_{0.42}$	$0.41^{0.07}_{0.11}$	0.29	False
IPOC ID	$-0.26^{0.03}_{0.06}$	$0.02^{0.01}_{0.01}$	$2.37^{0.13}_{0.18}$	$0.03^{0.00}_{0.00}$	0.00	True